

Smartphone Addiction Prediction Using Machine Learning

A Data Science Approach to Identify Smartphone Addiction Patterns

Module: Introduction to Data Science

Coursework: Classification Model Project

Dataset: Smartphone Usage and Addiction Analysis

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1. Introduction

Smartphone usage has become a common part of daily life, especially among students and young adults. People use smartphones for communication, social media, education, entertainment and work-related activities. However, excessive smartphone usage can affect daily routines, sleeping patterns, stress levels and academic or work performance.

This project focuses on predicting smartphone addiction using machine learning. The main aim of this project is to build a classification model that can predict whether a user is addicted to smartphone usage or not, based on their usage behaviour.

The dataset used in this project contains details such as age, gender, daily screen time, social media usage, gaming hours, work or study hours, sleep hours, number of notifications, app opens per day, weekend screen time, stress level and academic/work impact. These features help to understand different smartphone usage patterns.

The target variable used in this project is `addicted_label`. A value of 1 represents an addicted user, while a value of 0 represents a non-addicted user. In this project, machine learning models were trained and evaluated to identify the best model for predicting smartphone addiction.

2. Dataset Description

The dataset used for this project is based on smartphone usage and addiction behaviour. It contains information about users and their daily smartphone usage patterns. The dataset was selected because it is suitable for a classification task, where the main aim is to predict whether a user is addicted to smartphone usage or not.

The dataset was obtained from Kaggle, which provides publicly available datasets for data analysis and machine learning practice.

The original dataset contained 7500 records and 16 columns. These columns included both numerical and categorical variables. Some of the numerical variables were age, daily_screen_time_hours, social_media_hours, gaming_hours, work_study_hours, sleep_hours, notifications_per_day, app_opens_per_day and weekend_screen_time. The categorical variables included gender, stress_level, academic_work_impact and addiction_level.

The target variable used in this project is addicted_label. This column has two classes. A value of 1 represents an addicted user, while a value of 0 represents a non-addicted user. Because the target variable has two classes, this project is considered a binary classification problem.

The dataset also included identification columns such as transaction_id and user_id. These columns were not useful for prediction because they only represented unique records. Therefore, they were removed during the data cleaning stage. The addiction_level column was also removed because it was closely related to the target variable and could cause data leakage.

Overall, this dataset was suitable for the project because it contained useful behavioural features that can be used to identify smartphone addiction patterns.

Dataset Details Table

Item	Description
Dataset Name	Smartphone Usage and Addiction Analysis
Number of Records	7500
Number of Original Columns	16
Target Variable	addicted_label
Task Type	Binary Classification

3. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the structure, quality and patterns of the dataset before building the machine learning models. In this stage, the dataset was checked for missing values, duplicate records, target variable distribution, feature distributions and possible outliers.

Visualisations were used to understand how different variables are distributed and how they may be related to smartphone addiction. This helped to identify important patterns in the dataset before applying machine learning algorithms.

3.1 Target Variable Distribution

The target variable of this project is `addicted_label`. This variable shows whether a user is addicted to smartphone usage or not. A value of 1 represents an addicted user, while a value of 0 represents a non-addicted user.

The distribution graph shows that the number of addicted users is higher than the number of non-addicted users. This means the dataset is not perfectly balanced. Because of this, accuracy alone is not enough to evaluate the model, so F1-score was also considered during model evaluation.

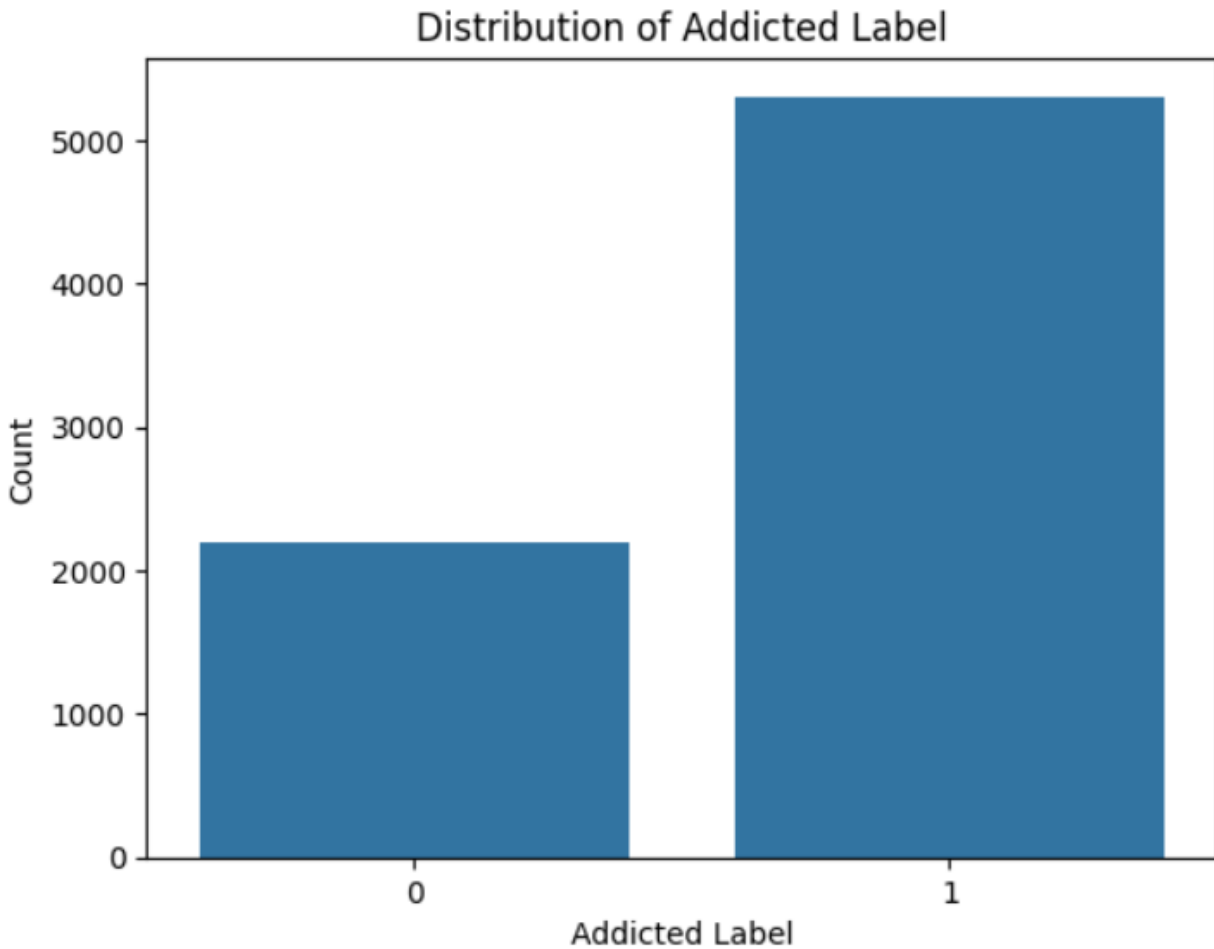


Figure 1: Distribution of Addicted Label

3.2 Gender Distribution

The gender distribution was analysed to understand the representation of users in the dataset. The dataset included different gender categories such as Male, Female and Other.

This analysis helped to check whether the dataset contained users from different gender groups. Having different gender categories makes the dataset more diverse and useful for analysing smartphone usage behaviour.



Figure 2: Gender Distribution of Users

3.3 Daily Screen Time Distribution

Daily screen time was analysed to understand how many hours users spend on smartphones per day. This is an important feature because users with higher screen time may have a higher chance of smartphone addiction.

The distribution graph shows the general pattern of daily smartphone usage in the dataset. It helps to identify whether most users have low, moderate or high daily screen time. This feature

was useful for understanding smartphone usage behaviour before building the machine learning models.

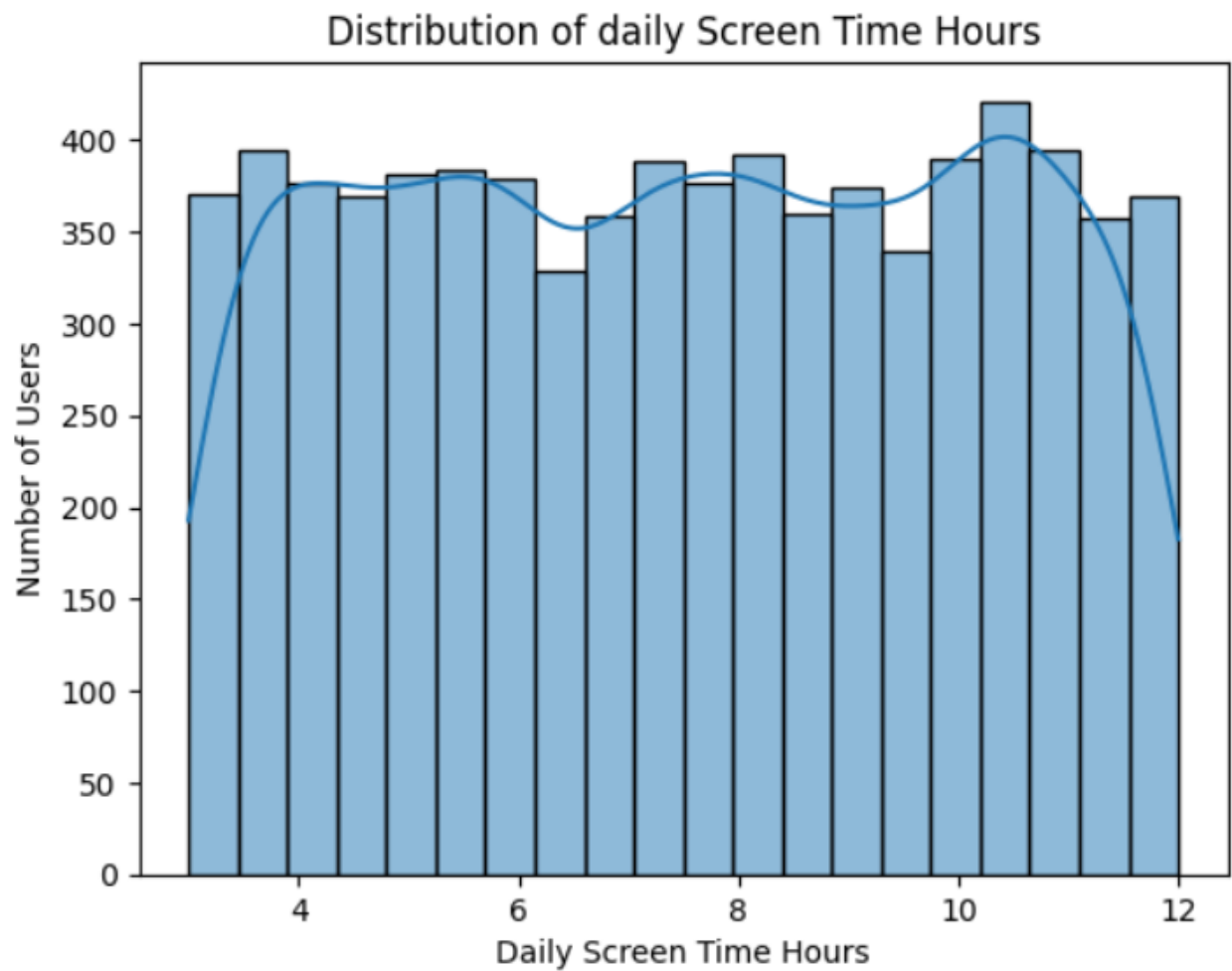


Figure 3: Distribution of Daily Screen Time Hours

3.4 Daily Screen Time by Addiction Label

Daily screen time was also compared with the addiction label to understand whether addicted and non-addicted users show different screen time patterns. This comparison is useful because screen time is one of the most important features related to smartphone usage behaviour.

The boxplot shows the daily screen time values for both classes of the target variable. Users labelled as addicted generally show higher daily screen time compared to non-addicted users. This suggests that daily screen time can be a useful feature for predicting smartphone addiction.

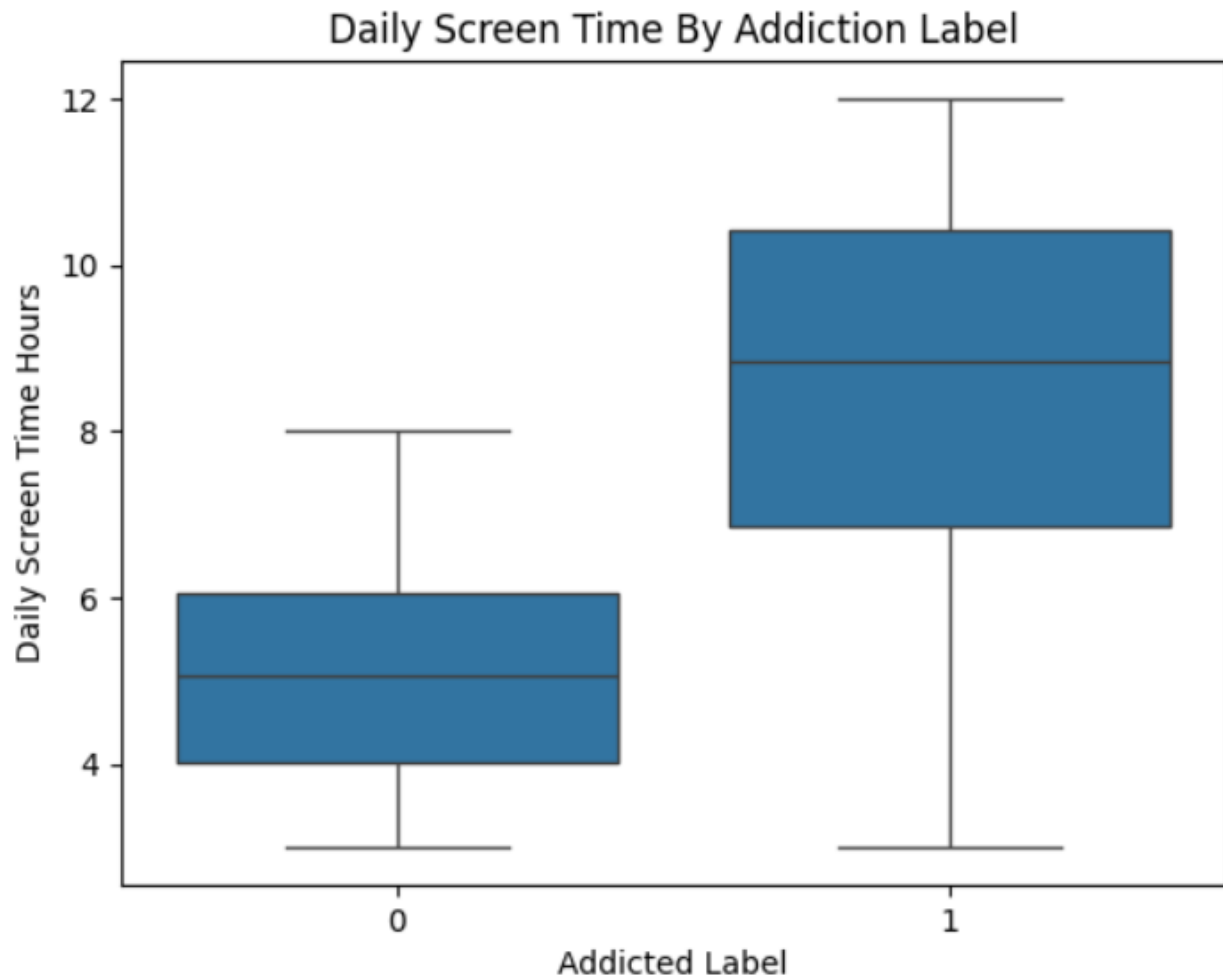


Figure 4: Daily Screen Time by Addiction Label

3.5 Weekend Screen Time by Addiction Label

Weekend screen time was analysed because users may spend more time on smartphones during weekends compared to regular weekdays. This feature helps to understand whether users with smartphone addiction show higher weekend usage patterns.

The boxplot compares weekend screen time between addicted and non-addicted users. Users labelled as addicted generally show higher weekend screen time compared to non-addicted users. This suggests that weekend usage behaviour may also be useful for predicting smartphone addiction.

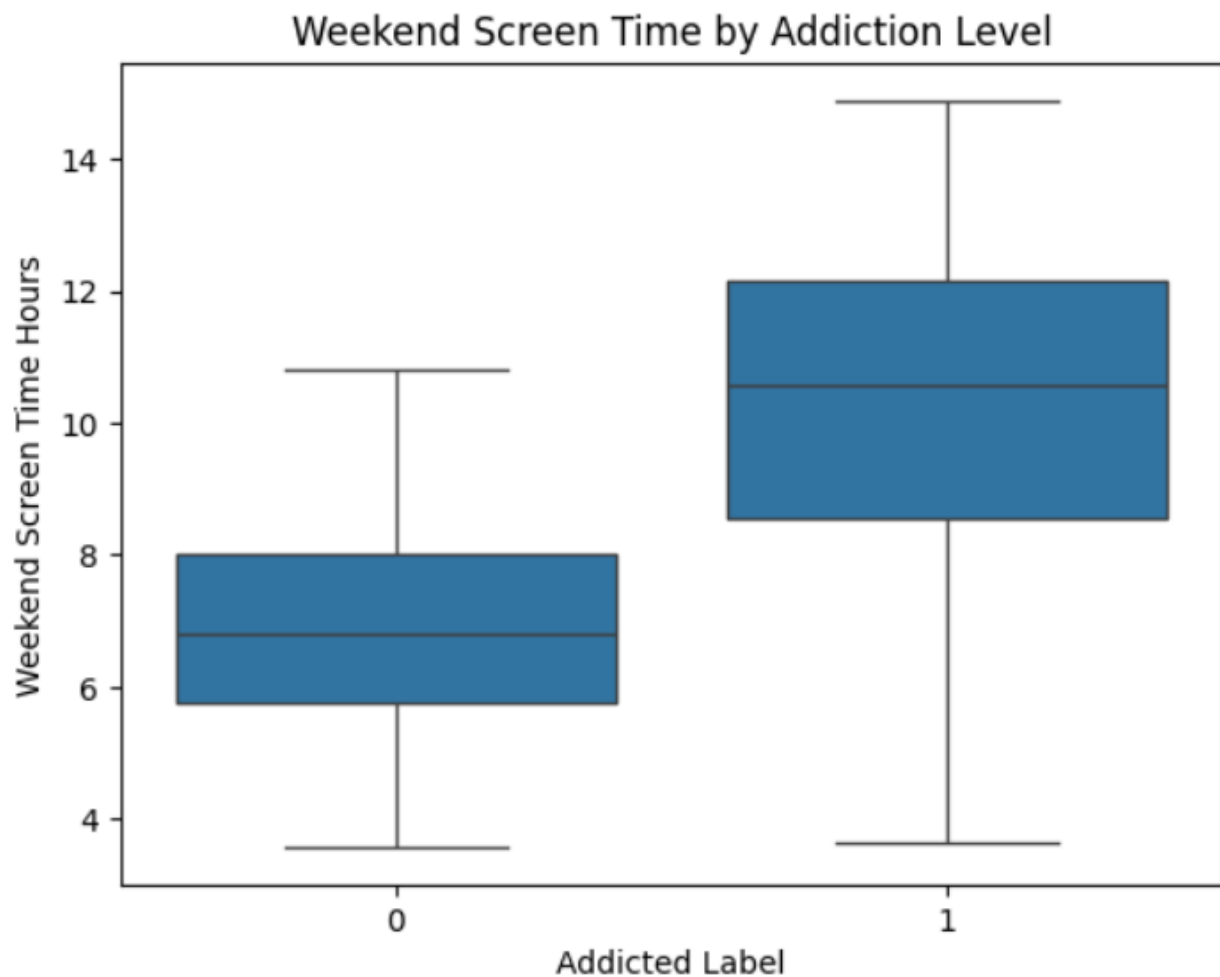


Figure 5: Weekend Screen Time by Addiction Label

3.6 Outlier Check

Outliers were checked to identify whether there were any unusual or extreme values in the numerical features. Boxplots were used for this purpose because they provide a simple visual way to identify values that are far away from the normal range.

In this project, outlier checks were done for numerical features such as daily screen time, sleep hours and weekend screen time. The boxplots helped to check whether the values in these columns were realistic and suitable for model building.

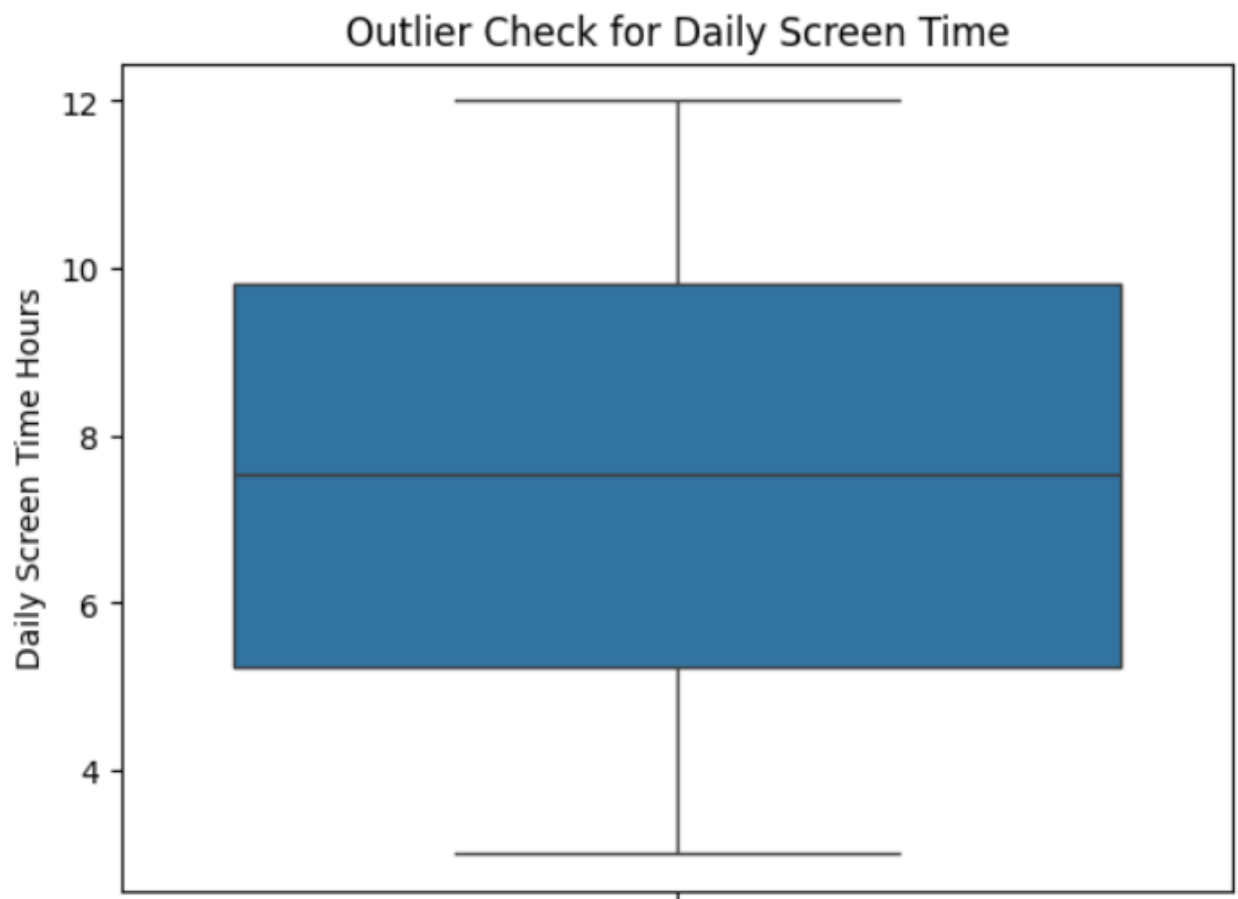


Figure 6: Outlier Check for Daily Screen Time

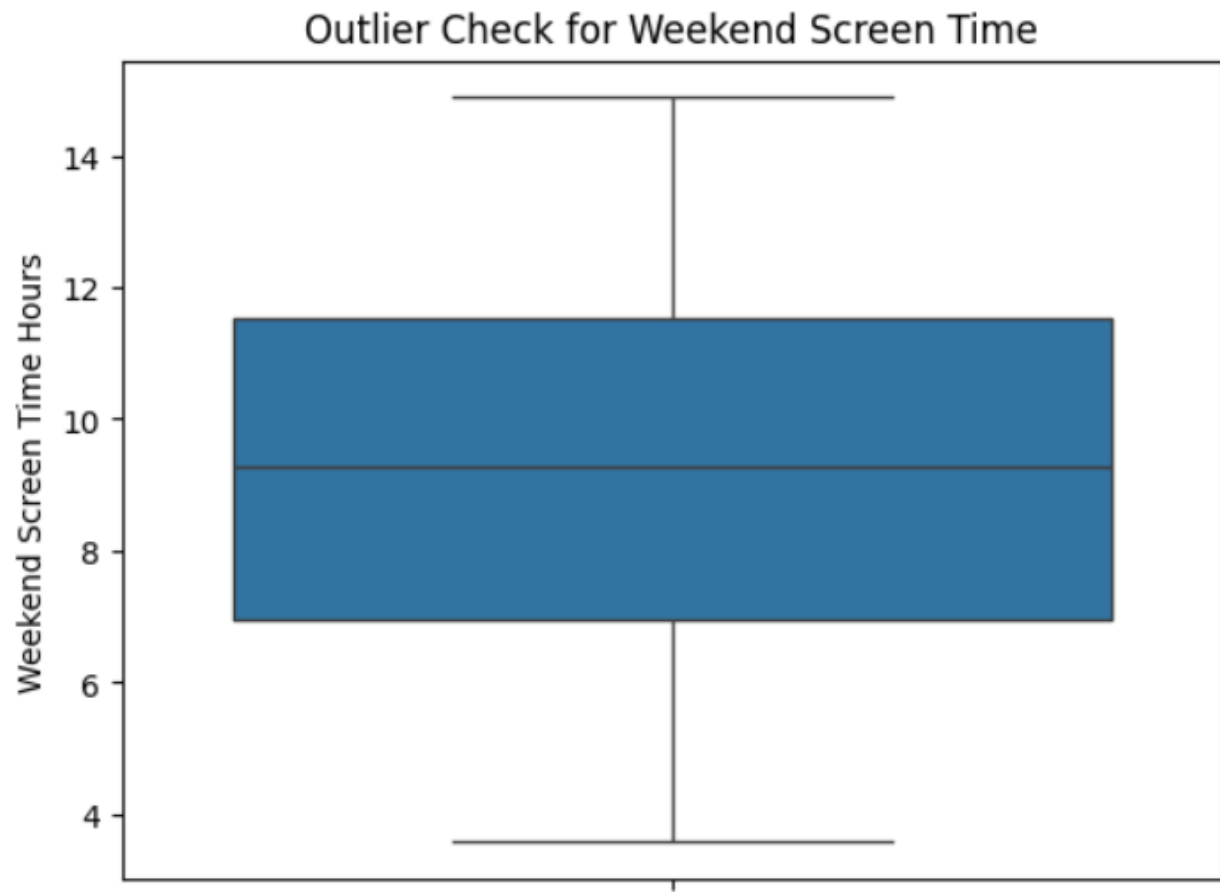


Figure 7: Outlier Check for Weekend Screen Time

Based on the boxplots, there were no major extreme values in these selected numerical features. Therefore, no special outlier removal was applied. The values appeared to be within a reasonable range for smartphone usage behaviour.

4. Data Cleaning and Preprocessing

Data cleaning and preprocessing were done before building the machine learning models. This step was important because machine learning algorithms require clean and properly prepared data to produce better results.

In this project, the dataset was checked for missing values, duplicate records and unnecessary columns. After that, categorical variables were converted into numerical form, and the dataset was divided into input features and the target variable. Finally, train-test splitting and feature scaling were applied before model training.

4.1 Missing Values and Duplicate Records

The dataset was first checked for missing values and duplicate records. Most columns did not contain missing values. The `addiction_level` column contained missing values, but this column was later removed because it could cause data leakage.

Duplicate records were also checked in the dataset. No duplicate rows were found, so no duplicate removal was required. This showed that the dataset was already in a good condition for further preprocessing.

4.2 Removing Unnecessary Columns

Some columns were removed from the dataset before model training. The `transaction_id` and `user_id` columns were removed because they were only identification columns and did not provide useful information for prediction.

The addiction_level column was also removed to avoid data leakage. This column was closely related to the target variable addicted_label. If this column was included in model training, the model could learn the answer too directly and give an unfairly high performance. Therefore, it was removed before building the models.

4.3 Encoding Categorical Variables

The dataset contained categorical variables such as gender, stress_level and academic_work_impact. Machine learning models cannot directly understand text values, so these categorical variables were converted into numerical form.

For this project, one-hot encoding was used to convert categorical variables into numerical columns. This made the dataset suitable for machine learning model training.

4.4 Train-Test Split and Feature Scaling

After preprocessing, the dataset was divided into input features and the target variable. The addicted_label column was used as the target variable, while the remaining columns were used as input features.

The data was split into training and testing sets using an 80:20 split. This means 80% of the data was used for training the models and 20% was used for testing. Stratified splitting was used to keep the class distribution similar in both training and testing data.

Feature scaling was also applied using StandardScaler. This was done because some numerical features had different value ranges. Scaling helps models such as Logistic Regression and KNN perform better.

Step	Description
Missing Values	Checked missing values in all columns
Duplicate Records	Checked duplicate rows in the dataset
Removed Columns	Removed transaction_id, user_id and addiction_level
Encoding	Converted categorical variables using one-hot encoding
Scaling	Applied StandardScaler before model training

5. Model Building

After completing data cleaning and preprocessing, machine learning models were built to predict smartphone addiction. Since the target variable `addicted_label` has two classes, this project was handled as a binary classification problem.

Two classification algorithms were used in this project: Logistic Regression and K-Nearest Neighbours. These models were selected because they are suitable for classification tasks and are easy to understand for a beginner-level machine learning project.

5.1 Logistic Regression

Logistic Regression was used as the first classification model in this project. It is suitable for binary classification problems because the target variable has two classes: addicted and non-addicted.

In this project, Logistic Regression was trained using the scaled training data. This model was used as a baseline model because it is simple, fast and easy to interpret. The model learned patterns from features such as screen time, social media usage, sleep hours and other smartphone usage behaviour to predict the addiction label.

5.2 K-Nearest Neighbours

K-Nearest Neighbours was used as the second classification model. KNN predicts the class of a new data record by looking at the nearest similar records in the training dataset.

In this project, KNN was trained using the scaled training data. The number of neighbours was set to 5. This means the model considered the five nearest records when making a prediction. KNN was useful for comparing model performance with Logistic Regression.

Model	Purpose	Reason for Selection
Logistic Regression	Predict addicted or not addicted users	Simple and suitable for binary classification
KNN	Predict addicted or not addicted users using similar records	Easy to understand and useful for comparison

6. Model Evaluation

After training the machine learning models, their performance was evaluated using different evaluation metrics. Evaluation is important because it shows how well the models can predict smartphone addiction on unseen test data.

In this project, accuracy, precision, recall, F1-score and confusion matrix were used to evaluate the models. Since the dataset was not perfectly balanced, F1-score was considered important together with accuracy.

6.1 Evaluation Metrics

Accuracy was used to measure the overall number of correct predictions made by the model. Precision was used to understand how many predicted addicted users were actually addicted. Recall was used to understand how many actual addicted users were correctly identified by the model. F1-score was used because it gives a balance between precision and recall.

The confusion matrix was also used to show the number of correct and incorrect predictions for each class. It helped to understand how the model performed for both addicted and non-addicted users.

6.2 Model Performance Comparison

The performance of Logistic Regression and KNN was compared using accuracy and F1-score. These two metrics helped to identify which model performed better for the smartphone addiction prediction task.

Model	Accuracy	F1-score
Logistic Regression	0.893333	0.925857
KNN	0.866000	0.907331

6.3 Confusion Matrix

In the Logistic Regression confusion matrix, 341 non-addicted users were correctly predicted as non-addicted, and 999 addicted users were correctly predicted as addicted. However, 97 non-addicted users were incorrectly predicted as addicted, and 63 addicted users were incorrectly predicted as non-addicted. This shows that the model made more correct predictions than incorrect predictions.

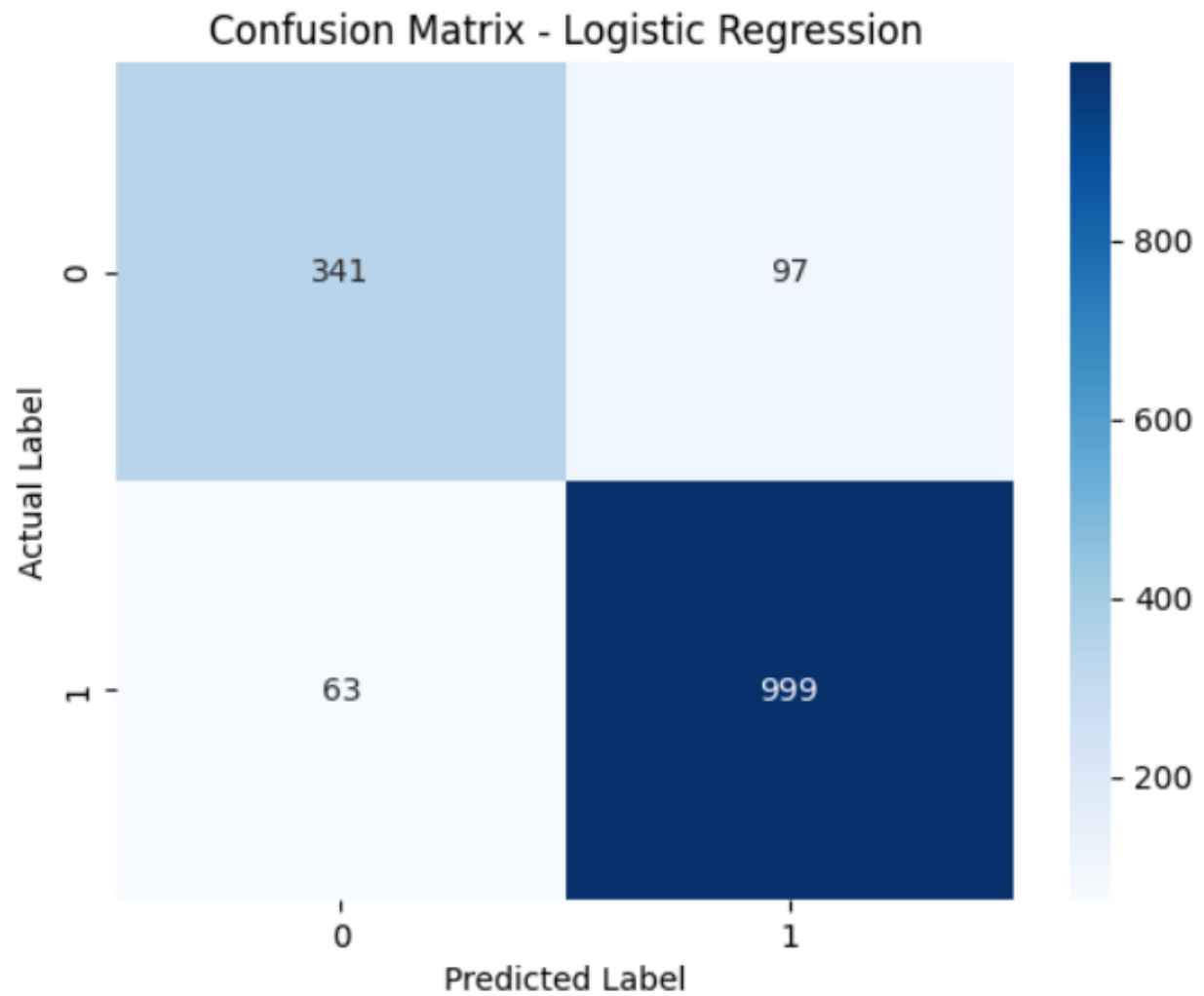


Figure 8: Confusion Matrix of Logistic Regression Model

6.4 Model Comparison Graph

A model comparison graph was used to visually compare the accuracy and F1-score of Logistic Regression and KNN. This made it easier to identify the model with better performance.

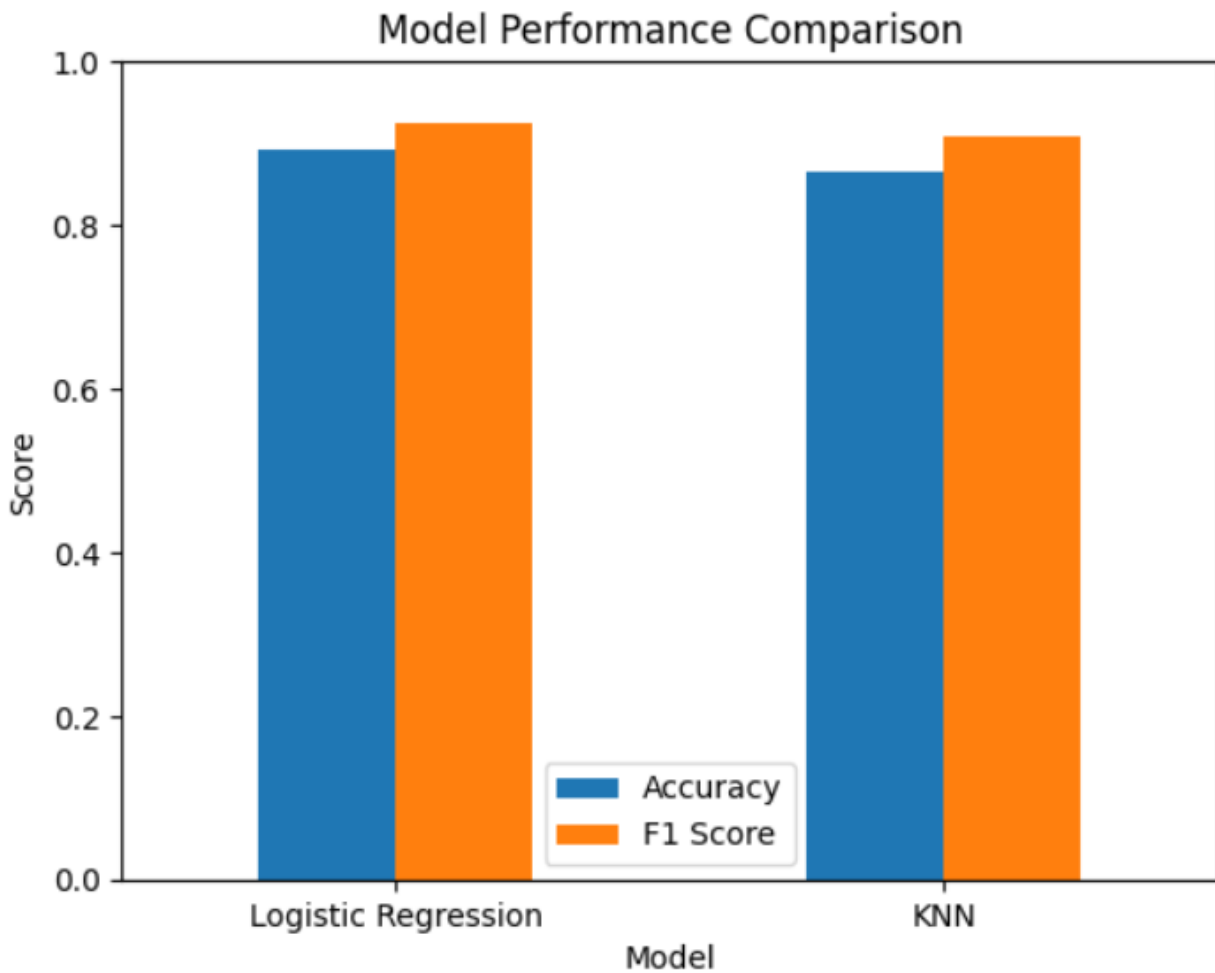


Figure 9: Model Performance Comparison

7. Result Interpretation

The results show that smartphone addiction can be predicted using user behaviour features. Features such as daily screen time, weekend screen time, social media usage, sleep hours, stress level and academic/work impact were useful for understanding smartphone usage patterns.

In this project, Logistic Regression and KNN were trained and evaluated. The models were compared using accuracy and F1-score. Since the target variable was not perfectly balanced, F1-score was considered important when selecting the best model.

The confusion matrix showed that the model correctly predicted many addicted and non-addicted users. However, some wrong predictions were also present. This shows that the model performed well, but it is not perfect.

Based on the model comparison, the model with the higher F1-score was selected as the best-performing model. This means that the selected model gave better balanced performance between identifying addicted and non-addicted users.

According to the model comparison results, Logistic Regression was selected as the best-performing model because it achieved higher accuracy and F1-score than KNN.

Overall, the results suggest that smartphone usage behaviour has a relationship with smartphone addiction. Higher screen time, weekend usage and social media usage can be useful indicators when predicting addiction patterns.

Finding	Interpretation
Target variable was imbalanced	F1-score was important together with accuracy
Daily screen time was higher for addicted users	Screen time is a useful predictor
Weekend screen time showed a clear pattern	Weekend usage may indicate addictive behaviour
Some wrong predictions were present	The model is useful but not perfect
Best model was selected using F1-score	Balanced performance was considered

8. Conclusion

This project was carried out to predict smartphone addiction using machine learning. The dataset contained smartphone usage behaviour details such as daily screen time, social media hours, gaming hours, work or study hours, sleep hours, notifications per day, app opens per day, weekend screen time, stress level and academic/work impact.

At the beginning of the project, the dataset was loaded and explored to understand its structure and quality. Basic checks such as shape, data types, missing values, duplicate records, target variable distribution and outlier checks were performed. These steps helped to understand the dataset before applying machine learning models.

Data cleaning and preprocessing were then carried out. Unnecessary columns such as transaction_id and user_id were removed because they were only identification columns. The addiction_level column was also removed to avoid data leakage. Categorical variables were converted into numerical form using one-hot encoding, and feature scaling was applied before model training.

Two classification models, Logistic Regression and KNN, were trained and evaluated. The models were compared using accuracy and F1-score, and Logistic Regression was selected as the best-performing model based on the evaluation results.

In conclusion, the project showed that smartphone usage behaviour can be used to predict whether a user is likely to be addicted or not. The model gave useful results, but it was not perfect because some incorrect predictions were still present. In the future, the project can be improved by using more data, adding more behavioural features and testing more machine learning algorithms.

9. References

The dataset used in this project was obtained from a publicly available source.

Dataset:

Smartphone Usage and Addiction Analysis Dataset.

<https://www.kaggle.com/datasets/guriya79/smart-phone>

Python Libraries Used:

- ❖ pandas
- ❖ numpy
- ❖ matplotlib
- ❖ seaborn
- ❖ scikit-learn

Tools Used:

- ❖ Google Colab
- ❖ Microsoft Word